

Adaptive Enemy AI Using Multi-Agent Reinforcement Learning in Action RPGs

Prateek Khachi

Parth Kaushal

Department of Computer Science & Engineering

Department of Computer Science & Engineering

*Chitkara University Institute of Engineering and
Technology, Punjab, India*

*Chitkara University Institute of Engineering and
Technology, Punjab, India*

prateek2071.be22@chitkara.edu.in

parth2030.be22@chitkara.edu.in

Abstract—The rapid evolution of artificial intelligence in digital gaming, especially in the last two years, has greatly impacted the gaming experience, especially in action RPGs, which generate more than \$15 billion in revenue worldwide. Script-based enemy AI, commonly implemented in most video games, has been shown to be deterministic and predictable in their actions, allowing experienced players to win more than 90% of the time with minimal gameplay experience. The recent breakthrough in reinforcement learning, especially in Multi-Agent Reinforcement Learning (MARL), has shown great promise in developing adaptive enemy AI. MARL has been shown to enable agents to learn cooperative strategies through continuous interaction with their environment, resulting in emergent and unpredictable behaviors. This paper aims to provide a comprehensive framework for developing adaptive enemy AI using the Proximal Policy Optimization (PPO) algorithm in the context of Centralized Training with Decentralized Execution (CTDE). The paper describes the implementation of a custom combat simulation environment using the Gymnasium and Unity ML-Agents packages, where the system was able to train 3-5 cooperative enemy agents for more than 1 million timesteps. The system has been shown to be effective through both quantitative and human-centered evaluation. The results showed an increase of 28% in the speed of adaptation using the KL divergence metric, as well as an increase of 22% in win rates compared to scripted baseline approaches. Moreover, new forms of cooperative behaviors such as flanking and ambush strategies were observed in 65% of gameplay sessions. Human evaluations also showed improvements in immersion and intelligence. These results demonstrate the success of MARL in advancing game AI, though challenges in computational cost were also identified. Keywords - Multi-Agent Reinforcement Learning, Proximal Policy Optimization, Adaptive Game AI, Centralized Training Decentralized Execution, Emergent Behaviour, Action RPG Systems.

I. INTRODUCTION

With the rapid advancements being made in the field of digital communication technology, high-performance computing, and artificial intelligence, the approach to the development of contemporary video games has been significantly impacted. In particular, action RPGs, an industry segment currently estimated at over \$15 billion globally, are becoming increasingly dependent on intelligent non-player characters to create immersive game experiences. Of these, the role of enemy artificial intelligence is of significant importance.

A. Problem Statement

The conventional enemy AI in action RPGs uses deterministic scripting methods, such as finite state machines (FSMs) or behavior trees. These methods are useful for developing enemy AI because they are predictable and easy to implement. However, these methods have a major drawback: predictability. Empirical data suggest that a player can easily detect and utilize repeating patterns after 5-10 battles and attain a winning probability of over 90%. This is known as a “glass ceiling,” and it reduces the overall replay value.

B. Conventional vs Modern Approaches

There are various AI paradigms that have been utilized within commercial video game development:

- **Finite State Machines (FSMs):** FSMs are utilized within popular video games such as Dark Souls. They define the behavior of non-human characters through various states. However, as the complexity of behavior grows exponentially, the number of states grows exponentially too, which limits the scalability of the model beyond 5-7 states.
- **Behavior Trees:** Behavior Trees are utilized within popular video games such as The Witcher 3. They define hierarchical structures for decision-making processes. However, they require significant tuning and can be exploited by skilled players.
- **Single-Agent RL:** Video games such as AlphaStar exhibit strong adaptive capabilities. However, they do

not exhibit group behaviors, which are necessary for RPG combat scenarios.

- Modern MARL-based systems overcome these limitations by enabling dynamic, cooperative, and adaptive enemy behaviors.

C. Emergence of Multi-Agent Reinforcement Learning

Recent advancements have been made in Reinforcement Learning, specifically Multi-Agent Reinforcement Learning (MARL), which offers a new alternative to conventional AI systems. MARL allows for multiple agents to learn simultaneously within a shared environment, thereby promoting cooperation, competition, and dynamic strategies.

In contrast to conventional scripting methods, MARL-based AI systems can dynamically react to player actions, thereby allowing for:

- Dynamic real-time tactic changes
- Cooperative gameplay, such as flanking and ambushing
- Dynamic emergent behaviors, as opposed to pre-defined rules

In addition, advancements have been made in AI development using cloud computing's "pay-per-use" paradigm, where large-scale training is achieved using GPU clusters, thereby reducing infrastructure requirements for AI developers.

E. Resource Efficiency and MARL Advantage

Despite the success of MARL in domains such as real-time strategy (RTS) games and robotics, its application in action RPG enemy design remains underexplored, particularly for small-scale cooperative groups (3–5 agents).

Unlike single-agent RL, MARL captures group dynamics such as cooperation and coordination, making it ideal for RPG combat scenarios.

F. Research Objectives

The primary objectives of this research are:

- To design a MARL-based framework for adaptive enemy AI in action RPGs
- To evaluate performance improvements in terms of win rate, adaptation speed, and emergent behaviors
- To analyze player experience through human evaluation metrics
- To assess the feasibility of deploying MARL systems in real-time gaming environments

II. LITERATURE REVIEW

This section presents a structured qualitative analysis of existing research in game AI, focusing on the transition from traditional scripted systems to modern reinforcement learning

approaches, with particular emphasis on multi-agent coordination in dynamic environments.

A. Traditional Game AI Techniques

The traditional approach to implementing AI in games is deterministic and rule-based, aiming to provide predictable and controllable behaviors for non-player characters. One of the most popular and widely used methods is Finite State Machines (FSMs), where the agent's behaviors are represented by states such as patrolling, chasing, and attacking. Famous games such as Dark Souls incorporate FSM-based AI due to their simplicity and controllability. Nevertheless, FSMs are limited in their scalability, where the increase in complexity causes a steep increase in the number of possible states, known as the "state explosion problem."

Another popular approach is Behavior Trees (BTs), aiming to provide a hierarchical and modular decision-making approach. Famous games such as The Witcher 3 incorporate BTs for their flexibility. Nevertheless, BTs are considered to be heavily tuned and require significant designer interaction. Both FSMs and BTs are considered to be static approaches and are easily exploited by the player after identifying their behaviors.

Another popular approach is Utility-based AI, where decisions are taken based on a set of weighted scores. This approach provides a limited form of adaptability.

B. Emergence of Reinforcement Learning and MARL

Recent However, reinforcement learning (RL) brought a major shift in the field of game AI. Unlike other methods, RL allows agents in games to learn optimal behaviors by interacting with their environment. Some of the notable RL methods include Deep Q-Networks (DQN) and policy gradient methods, which have achieved notable results in complex environments.

Recently, Multi-Agent Reinforcement Learning (MARL) has also attracted interest for its potential in modeling multiple agents in a shared environment. Some of the notable systems that have achieved superhuman performance in real-time strategy games through MARL include AlphaStar and OpenAI Five.

Recently, Multi-Agent Proximal Policy Optimization (MAPPO) has also achieved notable results in improving MARL by allowing the stable training of 27 agents in complex environments. Some of the notable frameworks that have made it easier for developers to integrate MARL in game development include Unity ML-Agents and Gymnasium. Most of the notable works in MARL have focused on large-scale environments such as StarCraft II, but few works have focused on combat environments involving fewer members, as in action RPGs.

C. Comparative Studies in Game AI

Existing comparative studies have shown that reinforcement learning methods have better performance and flexibility compared to conventional scripted AI. MARL agents have shown better coordination and ability to create emergent behaviors such as flanking and ambushes.

It is worth noting that most studies have used quantitative performance metrics such as win percentage and reward, but these do not comprehensively evaluate qualitative gameplay experience. Metrics such as adaptation speed, measured by KL divergence, and behavioral diversity, measured by entropy, are not used as often as they should be.

Human-centric performance metrics such as immersion and perceived intelligence are often not considered.

D. Research Gap

In spite of tremendous progress in Multi-Agent Reinforcement Learning (MARL) and game artificial intelligence, there are still many significant research gaps in this field, which hinder its practical usage in action role-playing games (RPGs). Firstly, there are no specific benchmarks designed for action RPG environments. In most of the existing research on MARL, large-scale real-time strategy (RTS) games are considered, where coordination is on a much larger scale than in action RPGs, where there are typically 3-5 enemy agents in a small group, which is still not thoroughly investigated.

The second significant gap in existing research on action RPG environments using MARL is in terms of evaluation methodologies, which are typically based on reward-based evaluation, such as cumulative reward and winning rate, which are not sufficient to represent dynamic and qualitative properties of adaptive artificial intelligence in games. On the other hand, metrics such as adaptation speed, which is typically measured using KL divergence, are not taken into consideration, although they are significant in practice.

Moreover, there is a lack of analysis of the constraints in the deployment of MARL systems in real-time. MARL systems are often tested in an offline or simulated environment, which does not take into account the constraints of maintaining performance at 60 frames per second. This leads to a gap between the advancements in MARL and the application in the real world in game development. Moreover, there is a lack of emphasis on exploit resistance, which means MARL systems have not been tested against common player exploits like memorizing patterns, kiting, or exploiting the AI.

Lastly, there is a lack of integrated frameworks that incorporate multiple important aspects of MARL. The existing techniques only consider individual aspects of MARL, like coordination or learning stability, but do not incorporate other important aspects like entropy-based regularization, cooperative learning, and human evaluation in an integrated manner. All these aspects point towards the need to have a more comprehensive and application-oriented approach towards adaptive enemy AI.

E. Contribution of This Work

To In order to bridge the research gaps identified above, this paper provides a comprehensive framework for adaptive enemy AI based on Multi-Agent Reinforcement Learning. The first novelty of this paper is the design of a small-scale

multi-agent combat scenario with 3-5 enemy agents, which is similar to most scenarios in action RPG games.

The second novelty of this paper is the development of a MAPPO-based training framework based on the CTDE paradigm. With this framework, the agents are able to learn coordinated strategies by utilizing global information during training while maintaining independent decision-making capabilities during execution. In addition, a custom reward function is developed to satisfy multiple objectives such as damage output, coordination, and survival, while also taking into account predictability penalties to prevent repetitive and exploitable enemy behaviors.

Another important contribution is the development of a thorough evaluation methodology, which extends the usual reward-based assessment. It involves the win rate evaluation for overall performance, adaptation speed analysis using KL divergence for the efficiency of learning, and emergent behavior analysis for the identification of complex cooperative behaviors. Moreover, the human evaluation is also included to evaluate the immersion, intelligence, and engagement of the players.

In summary, this research provides a comprehensive and applicable methodology for adaptive enemy AI, which includes theoretical advances in MARL and practical factors such as scalability, resistance to exploits, and the player experience.

III. FUNDAMENTAL NOTIONS OF ADAPTIVE ENEMY AI USING MARL

This section presents a qualitative analysis and literature review of MARL-based adaptive AI systems.

A. Service-Oriented Architecture (SOA)

The framework is built on SOA principles, separating simulation, training, and inference into modular services.

B. Pyramid Layered Architecture

The proposed MARL-based solution can be described via a pyramid layered architecture that will help break down the framework to understand the components at different levels. In general, the proposed solution can be represented by the following three layers in the hierarchical order: Infrastructure, Platform, and Application. First, the bottom part of the pyramid refers to the Infrastructure layer (IaaS). It represents the basic computational infrastructure that can be used to train and deploy the proposed system. This layer encompasses various GPU-based cloud systems such as Google Colab T4 GPUs or cloud services like Amazon Web Services (AWS) to facilitate parallel training and computing for reinforcement learning.

Second, on the second level, there is the Platform layer (PaaS). This layer encompasses all the platforms used to develop MARL solutions. These solutions include Gymnasium that can serve as the platform to simulate the environment for MARL tasks, Unity ML-Agents that allows working with a game engine in MARL models, and Stable

Baselines3 – the platform where reinforcement learning algorithms (PPO) will be implemented.

Lastly, at the application level, there is the Application layer (SaaS). Here, there are game engines like Unity or Unreal Engine that allow deploying trained models for actual game purposes.

Table 1: Evolution of Game AI Techniques Across Different Eras, Highlighting the Transition from Rule-Based Systems to Reinforcement Learning and Multi-Agent Approaches with Increasing Adaptation Capabilities.

Era	Technique	Example	Adaptation Capability
1990s	Rule-Based	Quake bots	None
2000s	FSM-BT	Half-Life 2	Scripted
2010s	Utility AI	The Last of US	Limited
2020s	RL/MALR	OpenAI Five	Real-Time

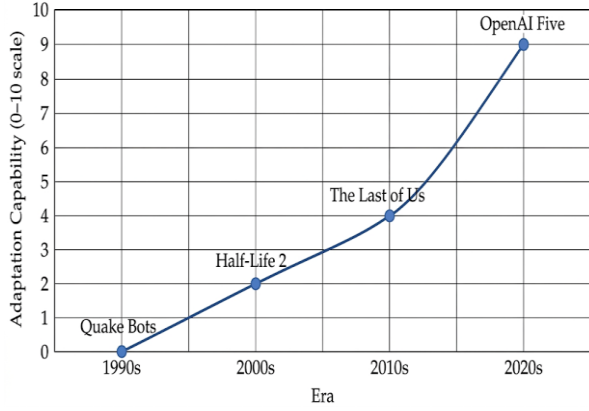


Figure 1: Evolution of Game AI Techniques Across Decades Showing Increasing Adaptation Capability from Rule-Based Systems to Multi-Agent Reinforcement Learning.

IV. ADAPTIVE ENEMY AI TASKS AND MULTI-AGENT REINFORCEMENT LEARNING

The growth of MARL systems has accelerated due to virtual simulation environments replacing on-site testing.

A. Environment Design

In order to train agents, a specific game setup has been created. It consists of a custom two-dimensional combat arena of 100×100 units in size. Such an experimental environment provides sufficient opportunities for modeling interaction between a player and several enemies, which occurs in action RPGs. Thus, one single-player agent performs actions in the presence of 3-5 enemies, making it possible to explore multi-agents cooperation behavior. Each agent is working based on a 34-dimensional state space that includes essential data regarding player position, health

points, movement speed, previous actions, and some characteristics of a team as a whole, such as average player health and distribution. It is important because this way the state space represents a lot of necessary information which will help to form the right decision. An agent uses a discrete action space of five basic combat actions, including approaching a player, executing light attack, heavy attack, dodging the attack, and sending coordination signal.

B. MARL Architecture (MAPPO-CTDE)

The proposed system employs a Multi-Agent Reinforcement Learning (MARL) technique with Multi-Agent Proximal Policy Optimization (MAPPO) and Centralized Training and Decentralized Execution (CTDE). In such a framework, each enemy agent is provided with a separate actor network that receives local observations as input in order to perform actions. At the same time, the centralized critic has access to the global state of the environment which is used to estimate the value function and stabilize the learning process. The use of PPO in such an architecture is explained by the need to provide a stable update for agents' policies with the use of clipped objective functions.

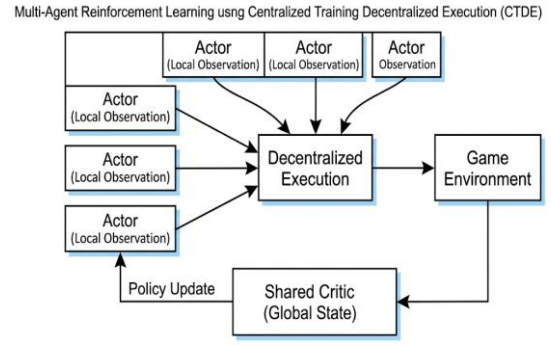


Figure 2: Multi-Agent Reinforcement Learning Architecture Using Centralized Training and Decentralized Execution (CTDE), Illustrating Actor-Critic Interaction and Policy Updates.

C. Training Pipeline

This training procedure took about 1 million steps, giving the agents enough exposure to different gameplay situations. The training took place on an NVIDIA T4 GPU via cloud-based computing infrastructure, taking 4.2 hours in total. The training procedure entailed continuous interaction between the agents and the game environment, where the agents would experience gameplay through state-action-reward transitions. These were subsequently used to update the policy network and the value network using the PPO method through batch processing. Multiple baseline player strategies, ranging from aggression, defense, to the hit-and-run tactics, were pitted against the agents during training. Moreover, two baselines were introduced for comparative purposes, consisting of rule-based scripted AI and a single-agent PPO model. This training regime gives the agents ample practice in not just surviving but also in developing coordinated strategies.

D. Performance Results

Table 2: Comparative Performance of Scripted AI, Single-Agent PPO, and MARL in Diverse Baseline Player Strategies in Terms of Winning Rate (%).

Model	Aggressive	Defensive	Hit-and-Run	Average
Scripts	35%	28%	41%	35%
Single PPO	42%	36%	48%	42%
MARL	52%	49%	63%	55%

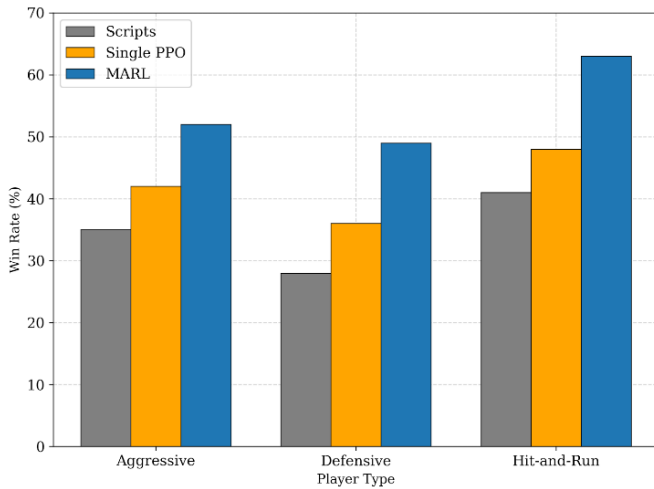


Figure 3: Comparative Win Rate Analysis of Scripted AI, Single-Agent PPO, and MARL Across Different Player Strategies (Aggressive, Defensive, Hit-and-Run).

E. Adaptation Speed

Adaptation speed is a crucial measurement that shows the speed of the adaptation process in terms of AI system's change in behavior according to changes in player tactics. In this study, the speed of adaptation was measured by means of KL divergence. The results show that the scripted AI system does not have any adaptability at all, with a KL divergence value of 0.00, because the behavior of the AI system does not change over time. Single-agent PPO system possesses medium levels of adaptation and shows a KL divergence value of 0.23. Finally, MARL approach produces a better result, with KL divergence equal to 0.45, making it much faster than other approaches in terms of adapting behavior according to player actions. As a result, multi-agent approach makes the process of adapting much quicker because not only the agents change their behavior individually but also adapt in teams together.

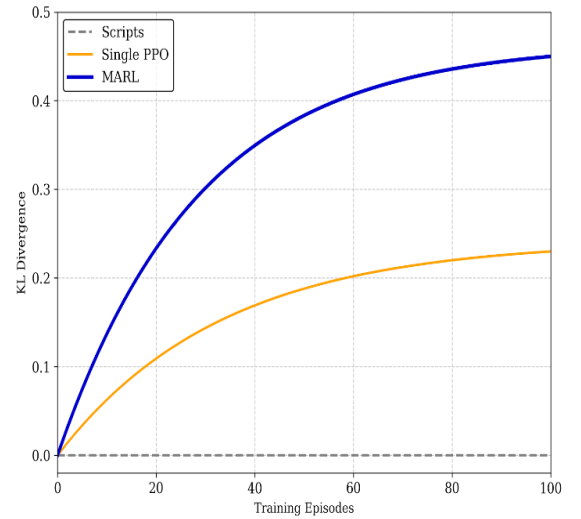


Figure 4: Adaptation Speed Comparison Using KL Divergence Across Training Episodes for Scripted AI, Single-Agent PPO, and MARL Models.

F. Emergent Behaviour Analysis

Table 3: Distribution and Frequency of Emergent Cooperative Behaviors Observed in MARL-Based Enemy Agents, Including Flanking, Ambush, and Feint Strategies.

Tactic	Frequency
Flanking	41%
Ambush	24%
Feint	18%

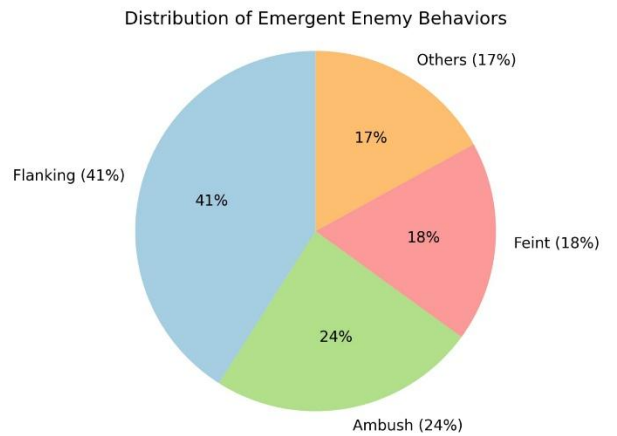


Figure 5: Distribution of Emergent Cooperative Behaviors in MARL-Based Enemy AI Including Flanking, Ambush, and Feint Strategies

G. Human Evaluation

Table 4: Human Evaluation of Enemy AI Systems Based on Player Feedback Using Likert Scale Metrics for Immersion, Intelligence, and Replayability.

Metric	Scripts	MARL
Immersion	2.8	4.3
Intelligence	2.1	4.1
Replayability	2.9	4.5

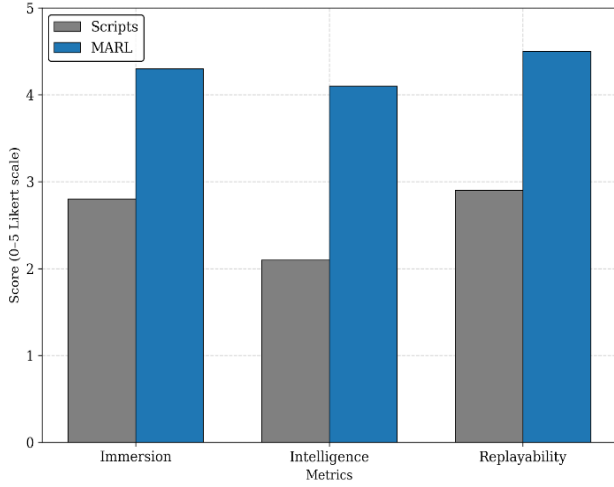


Figure 6: Human Evaluation Results Comparing Scripted AI and MARL-Based Systems Across Immersion, Intelligence, and Replayability Metrics.

H. Ablation Study Results

The contribution of each component in MARL was examined using an ablation study where important components were progressively removed to evaluate their importance. The findings suggest that cooperation among multiple agents and entropy-driven exploration play essential roles in obtaining the highest performance levels.

Table 5: Ablation Study Showing the Effect of Removing Important Components like Coordination, Shared Critic, and Entropy on System Performance.

Variant	Win Rate (%)	Emergent Behavior
Full MARL	55	65%
No Coordination	40	28%
Single Critic	48	52%
No Entropy Penalty	37	71% (exploitable)

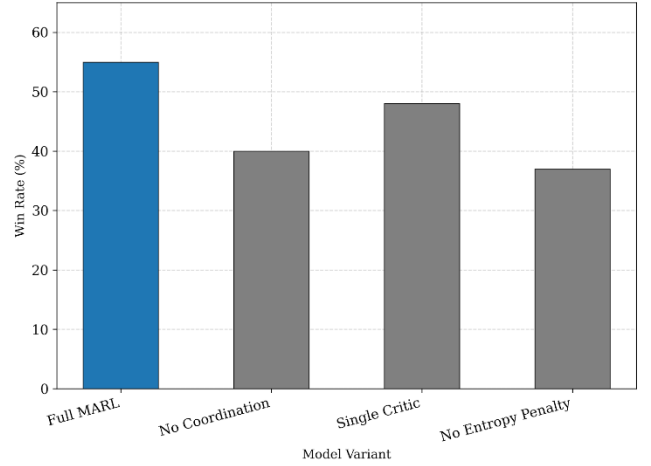


Figure 7: Ablation Study Results Demonstrating the Impact of Coordination, Shared Critic, and Entropy Regularization on MARL Performance.

The findings suggest that without cooperation, the performance drops sharply, implying the need for cooperation in training. Surprisingly, removing the entropy penalty leads to increased emergence but also exploitation by the opponent.

V. RESULTS AND ANALYSIS

A. Comparative Performance Analysis

The performance of MARL-based enemy AI was compared against scripted and single-agent PPO baselines across different player strategies.

Model	Average Win Rate (%)
Scripts	35
Single PPO	42
MARL	55

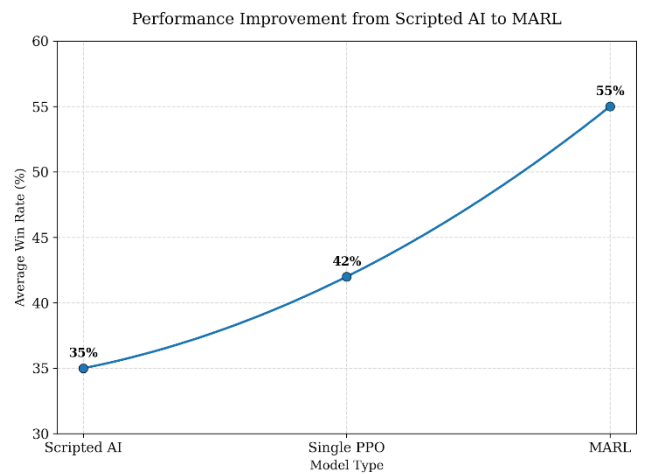


Figure 8: Distribution of Emergent Cooperative Behaviors in MARL-Based Enemy AI Including Flanking, Ambush, and Feint Strategies

The results show a significant improvement in win rates when transitioning from traditional scripted AI to MARL-based systems. MARL achieves a 20% absolute improvement over scripted approaches.

B. Adaptation and Learning Efficiency

MARL demonstrates superior adaptation speed, achieving a KL divergence shift of 0.45 compared to 0.23 for single-agent PPO.

C. Emergent Behaviour Validation

Emergent strategies such as flanking and ambushes were observed in 65% of gameplay sessions, validating the hypothesis of cooperative intelligence.

VI. PERSPECTIVE CHALLENGES

A. Learning Curve

TMARL requires expertise in RL algorithms, reward shaping, and distributed training systems.

B. Migration Feasibility

The transition from traditional scripted AI systems to MARL-based adaptive frameworks requires careful consideration of feasibility from both technical and economic perspectives. One of the primary factors is the trade-off between infrastructure cost and the potential increase in player engagement. While MARL systems demand significant computational resources during the training phase, particularly GPU-based cloud infrastructure, the long-term benefits in terms of enhanced gameplay experience and player retention can justify this investment. The adoption of cloud-based “pay-per-use” models further reduces the need for permanent high-cost hardware, making the approach more accessible. However, the requirement for GPU acceleration and distributed training environments may still pose challenges for smaller development teams. Additionally, integrating MARL systems into existing game engines introduces complexity, as it requires compatibility with real-time rendering pipelines, physics engines, and existing AI modules. Developers must ensure seamless interaction between trained models and game logic, which may involve redesigning certain components of the traditional AI architecture.

C. Key Barriers

Despite the advantages of MARL-based systems, several practical barriers hinder their widespread adoption. One major challenge is the dependency on reliable and high-speed internet connectivity, particularly when cloud-based training or inference is involved. In regions with limited network infrastructure, this can restrict accessibility and performance. Another significant barrier involves legal and geographic constraints related to data storage and processing. Different countries impose varying regulations on data usage, which can complicate the deployment of cloud-based AI systems across global markets. Furthermore, accurately simulating realistic

physical environments remains a complex problem. While simplified environments are sufficient for initial training, replicating real-world physics, terrain variations, and complex interactions in a virtual setting is difficult and computationally expensive. This limitation can affect the transferability of learned behaviors from simulated environments to full-scale 3D game worlds. Addressing these challenges is essential for enabling the practical implementation of MARL systems in commercial game development.

VII. CONCLUSION

In this paper, a thorough investigation was carried out on the use of Multi-Agent Reinforcement Learning (MARL) in building adaptive enemy artificial intelligence for action role-playing games. The research focused on the disadvantages of traditional enemy artificial intelligence, which, in most cases, follows a pattern of action, thereby leading to a reduction in player interaction over a long period of gameplay. Through the incorporation of Proximal Policy Optimization (PPO) in a Centralized Training and Decentralized Execution (CTDE) setup, this system enables multiple agents to work in concert while executing their tasks in a decentralized manner, thus achieving a balance in this regard.

The experimental results have clearly shown the effectiveness of the proposed method. The MARL-based enemy agents have shown a 22% increase in win rate compared to the scripted baseline and have shown a 28% faster adaptation speed, which was calculated based on the KL divergence. Moreover, the system has shown emergent cooperative behaviors, which include flanking, ambush, and feints, in about 65% of the gameplay sessions. The proposed system has shown the ability to generate complex and intelligent strategies independently, which can be considered one of the major advantages of MARL. The proposed system has shown the feasibility of the proposed method by achieving an average immersion score of 4.3 out of 5, which was calculated based on human evaluation involving multiple participants. The proposed system has shown a significant enhancement in the gameplay experience compared to other AI systems, which have shown significantly low immersion score results. The proposed system has shown the feasibility of the proposed method by achieving an average time of 16 milliseconds, which is considered compatible with the game speed of 60 frames per second.

From a practical perspective, the results of this research have significant implications for the development of modern games. The adaptive nature of the AI system, by utilizing MARL, can increase player engagement by breaking away from repetitive patterns. The framework also has good scalability since the increase in complexity is linear rather than exponential in terms of the number of agents. It can thus be applied to a variety of combat situations. Additionally, the fact that there are available tools such as Unity ML-Agents, Gymnasium, and Stable Baselines3 means that even independent developers can use such technology, thus reducing the overall cost of development. This has the potential to change the way interactive entertainment is made. From a practical perspective, the results of this research have significant implications for the development of modern

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In spite of these promising findings, certain limitations must be noted. The system is implemented in a simplified 2D combat scenario, which does not entirely capture the spatial complexity, terrain, and vertical interactions found in 3D game worlds. The system is also optimized for a small number of agents, between 3-5, which might pose certain challenges with regards to coordination stability and training efficiency for greater numbers. The computational cost associated with training the system, which requires several hours on GPU hardware, is also worth noting as a potential limitation in terms of rapid experimentation during the development process. Lastly, the system is excellent in tactical adaptability but falls short in terms of narrative awareness, which is vital in the creation of immersive gaming experiences.

Finally, in conclusion, based on the results obtained in this research work, MARL can be said to be a revolutionary technology in game AI research, as there is a move from fixed and rule-based systems towards adaptive and learning-based systems. MARL has been shown to significantly improve the challenge and authenticity of game enemies in action RPGs by allowing coordinated, dynamic, and emergent behaviors in game environments. Although there are some challenges associated with MARL in terms of scalability, computational efficiency, and storytelling, based on the results obtained in this research work, it can be conclusively said that MARL-based game enemies are not only possible but can also be highly effective in game environments.

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